$$J = \frac{|P \cap G|}{|P \cup G|}, \qquad J\&F = \frac{J + F}{2}$$

$$|P| = 12,\ |G| = 10,\ |P \cap G| = 8$$

$$J = \frac{8}{12 + 10 - 8} = \frac{4}{7} \approx 0.571$$